

UKHTA, Russian Federation

WMO: 236060

Lat: 63.5753N

Lon: 53.8058E

Elev: 449

StdP: 14.46

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.2	-27.0	-38.6	0.6	-33.2	-32.1	0.9	-26.8	17.6	17.3	15.6	12.3	2.8	20	0.705

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.7	82.2	65.9	78.4	64.3	74.4	62.0	68.3	78.2	66.0	74.8	63.9	72.1	5.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.7	93.2	71.8	62.7	86.6	70.1	60.5	80.0	68.5	32.9	78.4	31.0	74.8	29.4	72.3	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	15.0	13.8	DB	-35.4	87.2	6.8	4.1	-40.3	90.2	-44.2	92.6	-48.1	94.9	-53.0	97.9
			WB	-35.4	71.2	6.7	2.6	-40.2	73.0	-44.1	74.6	-47.9	76.0	-52.7	77.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	31.7	2.8	5.2	18.9	31.7	43.7	54.8	61.2	54.8	45.4	33.0	17.6	9.1
	DBStd	22.85	16.02	15.48	10.79	10.01	10.44	9.14	8.39	7.59	6.90	8.10	13.58	16.22
	HDD50	7524	1465	1255	965	551	258	52	10	33	169	528	971	1267
	HDD65	12262	1930	1675	1430	998	665	328	175	329	587	992	1421	1732
	CDD50	834	0	0	0	3	62	195	358	183	32	1	0	0
	CDD65	97	0	0	0	0	4	21	58	14	0	0	0	0
	CDH74	895	0	0	0	0	57	221	515	101	1	0	0	0
	CDH80	202	0	0	0	0	9	50	121	22	0	0	0	0
Wind	WSAvg	6.9	6.4	6.5	7.2	7.5	7.8	7.3	6.1	6.1	6.6	7.2	7.1	7.1
Precipitation	PrecAvg	22.9	1.4	1.1	1.3	1.3	1.8	2.8	2.4	2.7	2.4	2.3	1.8	1.5
	PrecMax	28.5	2.2	2.0	2.2	3.6	3.4	7.0	7.0	5.4	5.2	4.1	3.0	2.9
	PrecMin	18.7	0.7	0.4	0.2	0.1	0.3	0.9	0.7	0.5	0.9	1.2	0.4	0.9
	PrecStd	2.4	0.4	0.5	0.6	0.8	0.8	1.2	1.3	1.1	1.0	0.9	0.6	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.4	34.8	45.0	63.8	80.7	85.7	87.9	83.4	69.4	56.8	44.3	35.9
		MCWB	31.6	32.9	36.4	48.7	60.9	66.6	69.2	68.1	59.5	49.7	43.6	34.8
	2%	DB	30.0	32.2	38.5	54.6	73.7	80.3	83.4	76.1	63.0	50.4	37.4	33.3
		MCWB	29.5	30.8	32.7	44.0	57.2	64.4	66.9	63.9	55.1	47.5	36.7	32.3
	5%	DB	26.7	29.3	35.5	49.1	68.0	75.3	79.6	71.8	59.1	47.1	35.0	31.6
		MCWB	26.0	28.0	31.8	40.2	53.8	61.7	65.4	61.6	53.7	44.9	34.3	30.9
	10%	DB	23.1	26.2	33.1	44.9	62.2	70.4	76.1	67.5	55.6	44.0	32.8	29.0
		MCWB	22.1	25.0	30.6	37.9	50.4	58.8	64.0	59.8	51.8	42.4	32.1	28.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.8	33.4	37.3	50.1	62.3	69.2	71.7	69.4	60.6	51.8	42.9	34.8
		MCDB	32.2	34.4	43.2	61.8	74.1	79.4	81.3	79.2	67.9	55.3	43.8	35.2
	2%	WB	29.1	30.8	34.2	44.4	58.4	66.2	69.4	66.0	56.9	47.7	36.5	32.3
		MCDB	29.6	31.9	36.9	53.0	71.8	76.0	79.8	73.9	61.1	50.0	36.9	32.8
	5%	WB	25.9	27.9	32.4	41.4	54.7	63.5	67.2	63.4	54.5	44.9	34.1	30.7
		MCDB	26.6	29.0	34.8	47.1	66.2	73.1	76.3	69.4	58.4	46.8	34.5	31.4
	10%	WB	22.1	25.0	30.7	38.9	51.6	60.4	65.2	60.8	52.1	42.2	32.2	28.0
		MCDB	23.0	26.2	33.2	43.5	60.9	68.8	73.9	65.8	55.4	43.7	32.6	28.9
Mean Daily Temperature Range	5% DB	MDBR	11.8	12.8	15.5	15.4	17.5	17.3	17.7	15.0	11.7	7.7	9.0	10.4
		MCDBR	12.4	9.1	14.5	20.9	26.0	23.3	23.1	21.4	18.1	11.4	6.9	8.7
		MCWBR	12.4	8.6	11.0	13.2	14.1	11.6	10.4	10.7	11.1	9.2	7.1	8.9
	5% WB	MCDBR	12.1	9.2	11.8	17.6	24.3	21.0	21.0	18.8	15.7	10.5	6.5	8.4
		MCWBR	12.1	9.1	9.7	12.2	14.2	11.7	10.8	10.9	10.9	9.0	6.8	8.6
Clear-Sky Solar Irradiance	taub	0.236	0.285	0.320	0.352	0.354	0.362	0.410	0.377	0.337	0.307	0.235	0.226	
	taud	1.929	2.038	2.055	2.105	2.383	2.440	2.306	2.381	2.470	2.318	1.996	1.915	
	Ebn at Noon	104	193	237	258	272	272	252	252	240	193	105	43	
	Edh at Noon	10	22	33	39	33	32	35	30	23	17	9	5	
All-Sky Solar Radiation	RadAvg	62	284	726	1276	1590	1644	1648	1110	597	227	68	25	
	RadStd	6	40	90	137	133	144	148	128	80	29	15	3	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	+1.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page